PROJECT REPORT

(Project Term January - June 2025)

(Jewellery Shop Website)

Submitted by

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Course Code : INT221
Course Title : MVC Programming

Submitted To

(Kuldeep Kumar Kushwaha)

School of Computer Science and Engineering

DECLARATION

We hereby declare that the project work entitled ("Jewellery Shop Website") is an authentic record of our own work carried out as requirements of Capstone Project for the award of B.Tech degree in B-Tech CSE from Lovely Professional University, Phagwara, under the guidance of (Kuldeep Kumar Kushwaha), during January to June 2025. All the information furnished in this project report is based on my own intensive work and is genuine.

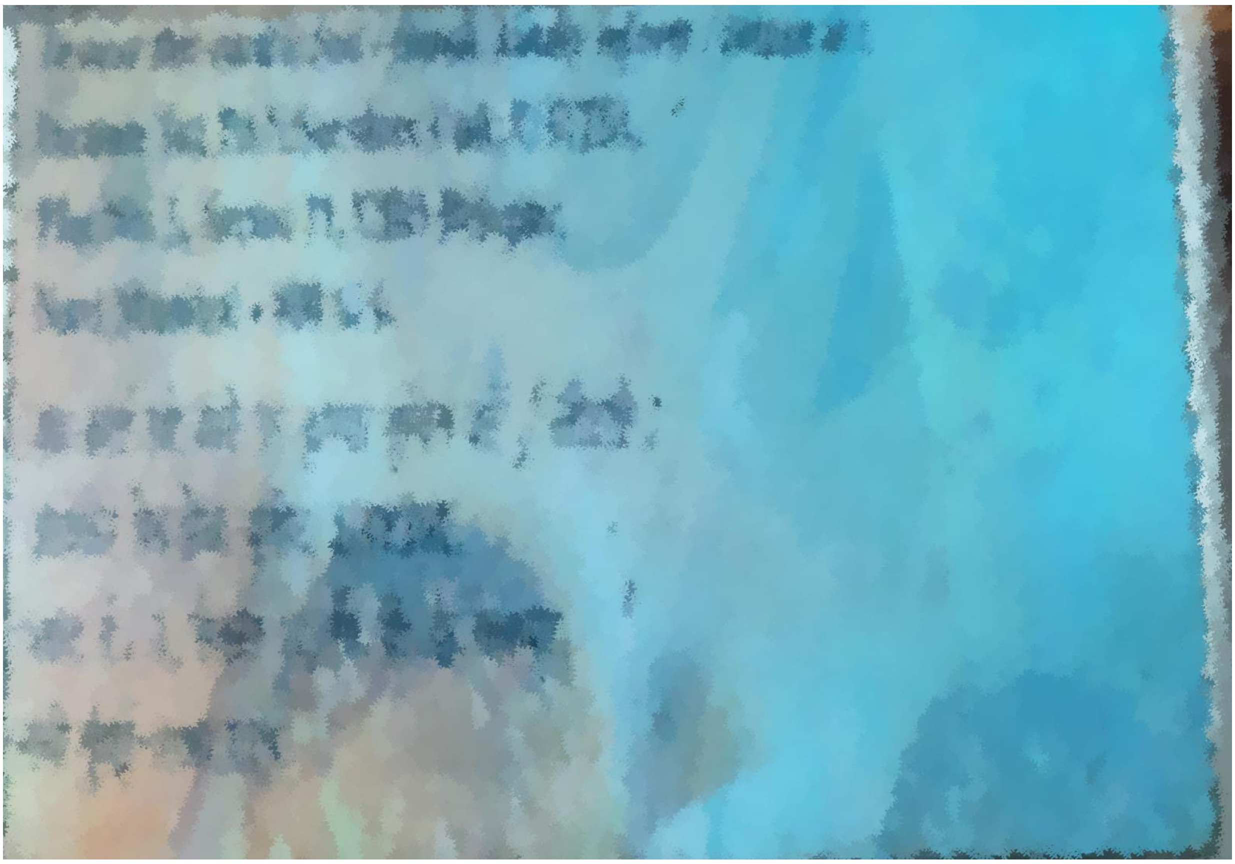
Name of Student: Tarun Choudhary
Registration Number: 12218125

TARUN

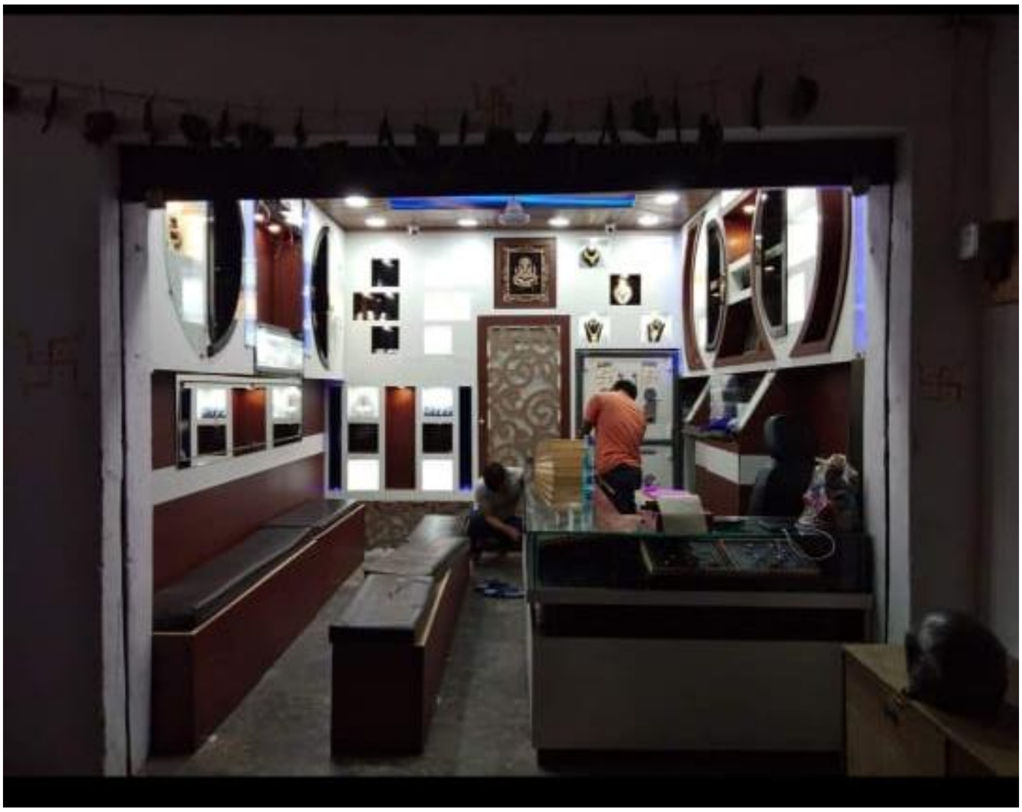
Date: 10-11-2024

Client Details

GST No :- 02ADUPV8110F1ZF



Location of Client shop



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1. INTRODUCTION

Jewellery Shop Website

In today's digital age, convenience and accessibility are crucial, especially for customers shopping for jewellery online. With busy schedules, customers need an intuitive platform that allows them to explore, select, and even customize jewellery items from the comfort of their homes. I propose a modern jewellery shop website that offers a seamless, user-friendly experience for browsing, purchasing, and even pre-ordering jewellery items.

The online jewellery shop will provide an engaging and personalized shopping experience:

- **Browse a Wide Selection**: Users can explore various categories like rings, necklaces, bracelets, and earrings. Each item will be presented with high-quality images, detailed descriptions, and customizable options such as size, gemstone, and metal type.
- **Customization Options**: For a more personalized touch, users can choose customization features like engraving or combining elements from different designs to create a unique piece.
- **Pre-order & Reservations**: The system will allow users to reserve high-demand items or place pre-orders for new collections before they launch. This feature is ideal for exclusive pieces and limited-edition items.
- **Flexible Payment Options**: Users can select their preferred payment method, either online (credit/debit cards, digital wallets) or in-store at the time of pickup.
- **User Account & Order Tracking**: Users can create accounts to save their favourite items, view purchase history, and track their orders.

Technology Stack:

- **Frontend**: HTML, CSS, JavaScript,Typescript
- **Backend**: PHP, Laravel Node.js, Express.js
- **Database**: MYSQL
- **Additional Libraries**: Bootstrap for responsive design, and jQuery where needed

This jewellery shop website will not only streamline the shopping experience but also increase user satisfaction by combining convenience, customization, and technology.

1.1. Purpose

Purpose:
This document outlines the requirements for developing an online jewellery shop website aimed at providing a seamless, convenient, and personalized shopping experience. In today's fast-paced world, the modern customer values convenience, ease of use, and a variety of choices when shopping for jewellery. This online platform will cater to these needs by allowing customers to browse, select, and even customize jewellery pieces from the comfort of their homes.

With a diverse range of jewellery items, from everyday pieces to luxury collections, this online jewellery shop will cater to users with different preferences and budgets, helping to transform the shopping experience by removing the need to visit physical stores and making high-quality jewellery accessible at any time.

Objectives:
The primary goal of this platform is to offer a user-friendly, visually engaging, and flexible online shopping experience. Here’s how it aims to enhance the jewellery shopping journey:

1. **Eliminating In-Store Waits and Hassles**:
   With this online platform, customers no longer need to visit a physical store and wait for assistance to view pieces. By browsing and purchasing jewellery online, they can enjoy a personalized shopping experience, free from queues and crowds, while exploring all available items at their own pace.

2. **Flexibility and Convenience for Busy Customers**:
   Whether browsing for a quick gift or carefully selecting an investment piece, customers can filter jewellery by type, price, metal, gemstone, and other specifications. Additionally, pre-ordering exclusive pieces or requesting customizations is made simple, allowing customers to secure their desired items in advance.

3. **Detailed Catalog and Customization Options**:
   Customers can browse a catalog of items including rings, necklaces, bracelets, and earrings, each with detailed images, descriptions, and specifications. For personalization, users can select engraving options or customize metal and gemstone choices. This enhances the uniqueness of each purchase, appealing to customers who value distinct, individualized jewellery.

Process:
The website is designed to be intuitive and user-friendly, ensuring a smooth experience:

- **Browsing and Selection**:
  Users can easily access the platform from any device and explore categories with clear visuals, item details, and prices. Enhanced product views and zoom options provide a close-up experience to inspect each piece before deciding.

- **Pre-ordering and Custom Requests**:
  For exclusive items, users can place pre-orders or make custom requests directly through the platform. This feature is ideal for limited-edition collections, ensuring customers can secure items that align with their unique preferences and style.

- **Checkout and Payment Flexibility**:
  The platform supports multiple payment methods, including credit/debit cards, digital wallets, or cash on pickup (for local orders), allowing customers the freedom to pay as per their convenience. Online payments will be secure and quick, with offline payment options available at the time of in-store pickup if required.

Features:

- **Account Management**: Users can create accounts to save favourite items, view order history, and track order statuses.

- **Personalized Recommendations**: Based on browsing history and preferences, users receive tailored product suggestions.

- **Order Tracking and Notifications**: Once an order is placed, customers will receive notifications on the status and estimated delivery or pickup time.

Technology Stack:

- **Frontend**: HTML, CSS, JavaScript, React.js, TypeScript

- **Backend:** PHP , Laravel Node.js, Express.js

- **Database**: MYSQL

This jewellery shop website will elevate the traditional shopping experience by blending luxury with the convenience of technology, providing a robust, customizable, and enjoyable platform for customers.

1.2 Scope

The jewellery shop website will allow users to browse collections, select items, request customizations, and make payments either in advance or at pickup. It will also provide shop owners with tools to manage orders, update inventory, and track customer preferences.

[1] User Convenience:

- Users can browse through a wide selection of jewellery items, including rings, necklaces, bracelets, and earrings, catering to a range of styles and occasions.
- The system allows users to explore and purchase jewellery online, providing a convenient, time-saving alternative to visiting a physical store.

[2] Efficient Shopping Process:

- The platform offers a simple yet effective purchasing process, allowing users to select items and request customizations, such as engravings or metal type, for a personalized experience.
- Users have the flexibility to choose between shipping to their address or picking up in-store, catering to their preferred shopping experience.

[3] Seamless Integration of Technology:

- The jewellery shop website seamlessly integrates technology to enhance user satisfaction and engagement.
- It leverages an intuitive platform that provides high-quality visuals and detailed descriptions, ensuring a smooth and enjoyable browsing and purchasing experience.

[4] Flexible Payment Methods:

- The website offers flexibility in payment options, allowing users to pay online through credit/debit cards or digital wallets or to pay in-store at pickup.
- This flexibility ensures customers can choose their preferred payment method, enhancing convenience and satisfaction.

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1.3 Definitions, Acronyms, and Abbreviations

- **Jewellery Shop Website**: A digital platform that allows users to browse, customize, and purchase jewellery items online, with options for in-store pickup or delivery.
- **User Convenience**: Refers to the ease and comfort experienced by users while interacting with the jewellery shop website, including browsing collections, selecting items, and making payments.
- **Shopping Experience**: The overall experience of exploring and purchasing jewellery, encompassing factors such as product quality, visual presentation, and ease of use.
- **Customization**: The option for users to personalize jewellery items, such as choosing specific metal types, gemstones, or engraving text.
- **User Satisfaction**: The degree to which users are content with their interactions and experiences with the jewellery shop website, including aspects such as product variety, customization options, and service quality.

Acronyms and Abbreviations:

- **SRS**: Software Requirements Specification
- **UI**: User Interface
- **UX**: User Experience
- **API**: Application Programming Interface
- **TBD**: To Be Determined
- **POS**: Point of Sale
- **PCI DSS**: Payment Card Industry Data Security Standard
- **HTTP**: Hypertext Transfer Protocol
- **HTTPS**: Hypertext Transfer Protocol Secure
- **SKU**: Stock Keeping Unit
- **CMS**: Content Management System

1.4 <u>Overview</u>

- ➢ The SRS contains detailed requirements for the development of Jewelry website.
- ➢ It is organized into sections such as Introduction, Functional Requirements, Non-Functional Requirements, Constraints, and Glossary.
- ➢ The Introduction section provides background information and the purpose of the document.
- ➢ Functional Requirements specify the actions the system must perform.
- ➢ Non-Functional Requirements specify the quality attributes of the system.
- ➢ Constraints outline any limitations or restrictions on the system's design and implementation.
- ➢ The Glossary defines key terms and acronyms used throughout the document.

2. General Description

The jewellery shop website is a stand-alone digital platform aimed at enhancing the shopping experience for jewellery customers. The primary objective is to offer a convenient and accessible way for users to explore, customize, and purchase a variety of jewellery pieces without needing to visit a physical store. Imagine a scenario where a customer is searching for a special piece of jewellery for an upcoming event but has limited time to shop in person. Traditional in-store shopping can be time-consuming, with limited product availability and personalization options. By providing a seamless online solution, this jewellery shop website aims to remove these barriers, allowing customers to browse collections, select items, and even customize jewellery pieces at their convenience. This streamlined experience makes jewellery shopping more accessible, personalized, and efficient.

Product Perspective
The jewellery shop website is an independent digital platform designed to enhance the shopping experience for customers. Its purpose is to provide a convenient and engaging way for users to browse, customize, and purchase jewellery online, offering a wide selection and personalized options.

2.2 Product Functions

- Allow users to browse a variety of jewellery items, including rings, necklaces, bracelets, and more.
- Enable users to select items, request customizations (such as engravings), and place orders for in-store pickup or delivery.

- Provide shop owners with a dashboard to manage inventory, update product listings, and track orders.
- Facilitate secure online payments and integration with various payment gateways.

2.3 User Characteristics

The primary users of the product include jewellery customers from diverse demographics, ranging from occasional shoppers to collectors and gift buyers. Users may have varying levels of familiarity with online shopping, but they all share a common need for convenient, accessible, and secure shopping for jewellery pieces.

2.4 General Constraints

- Compliance with data privacy and payment security standards (e.g., PCI DSS).
- Compatibility with various devices and browsers to ensure accessibility.
- Availability of secure internet connectivity for users accessing the platform.
- Adherence to budget and timeline constraints for development and deployment.

2.5 Assumptions and Dependencies

- Assumption that the shop will provide accurate and timely updates on product availability and specifications.
- Dependency on internet connectivity and compatible devices for users to access the website.
- Assumption that users will provide valid contact information for order tracking and communication.

3. Problem Analysis

3.1 Product Definition

The jewellery shop website aims to address the following challenges faced by jewellery customers and shop owners:

1. **Limited Access to Product Variety**: Customers may not have access to the full range of jewellery products in physical stores, limiting their ability to view and choose from a variety of designs.

2. **Time Constraints**: Customers with busy schedules may struggle to find time to visit physical stores, making it difficult to explore jewellery options in person.

3. **Inconvenience in Customization**: Customizing jewellery items (such as engraving or selecting specific materials) can be a time-consuming process in-store, often requiring multiple visits.

4. **Shopping Inconvenience**: Some customers may find it inconvenient to travel to jewellery stores, particularly those located far from their residence, or to shop during regular business hours.

5. **Limited User Experience**: Physical stores may not offer the level of personalization, detailed product views, or user-friendly options that can be provided on a digital platform, potentially leading to a less engaging shopping experience.

3.2 Feasibility Analysis

The feasibility of implementing the jewellery shop website is evaluated based on technical, operational, and economic factors:

1. **Technical Feasibility**:
    - Developing a user-friendly web platform with a secure and interactive interface for browsing, customizing, and purchasing jewellery items.
    - Integrating with payment gateways and other third-party services for a seamless shopping experience.
    - Ensuring scalability to handle multiple users and product updates as the catalog expands.
2. **Operational Feasibility**:
    - Assessing user adoption and the need for minimal training to ensure ease of use for customers with varying technological proficiency.
    - Providing ongoing customer support to assist users with product inquiries, order tracking, and customization requests.
    - Ensuring that shop owners can easily update inventory, manage orders, and track customer preferences within the platform.
3. **Economic Feasibility**:
    - Analysing the costs associated with developing, deploying, and maintaining the website to determine whether the project is financially viable.
    - Considering potential revenue streams, such as increased sales from online customers and reduced overhead from in-store operations.
    - Ensuring that the project is sustainable over the long term, with a potential for growth and adaptation as business needs evolve.

3.3 Project Plan

The project plan outlines the tasks, timelines, and resources required to develop and implement the jewellery shop website:

1. **Project Scope**: Define the scope of the project, including specific features and functionalities of the jewellery shop website, such as browsing collections, customization options, secure checkout, and inventory management.
2. **Requirements Gathering**: Collect requirements from stakeholders, including potential customers and shop owners, to ensure the platform meets their needs and preferences, such as customization capabilities and secure payment options.
3. **System Design**: Design the website architecture, user interface, and database schema for the jewellery shop platform based on gathered requirements. This includes developing an intuitive user experience for browsing, selecting, and customizing jewellery items.
4. **Development**: Develop the frontend and backend components of the jewellery shop website using appropriate technologies and frameworks, such as Laravel , PHP , TypeScript, Node.js, MongoDB, HTML, CSS, and JavaScript .
5. **Testing**: Conduct comprehensive testing to identify and resolve any bugs or usability issues before launch. This includes testing for cross-browser compatibility, performance, and security to ensure a seamless shopping experience.
6. **Deployment**: Deploy the jewellery shop website on a secure hosting solution or cloud infrastructure, ensuring accessibility to users across multiple devices and regions.
7. **Training and Support**: Provide training and support to shop owners on how to manage product listings, handle orders, and update inventory on the website, as well as support for customers in using the platform effectively.
8. **Monitoring and Maintenance**: Continuously monitor the website's performance and usage after deployment, providing regular updates, security patches, and customer support to ensure the website remains reliable and up-to-date.

Software Requirement Analysis

This section outlines the detailed requirements that will guide the design, implementation, and testing of the jewellery shop website. Each requirement is designed to be correct, traceable, unambiguous, verifiable, prioritized, complete, consistent, and uniquely identifiable.

4.1 External Interface Requirements

4.1.1 User Interfaces

- The user interface should be intuitive and easy to navigate for all user levels.
- Users should be able to browse jewellery collections, view product details, customize items, and place orders with ease.

4.1.2 Hardware Interfaces

- The system should be accessible via standard web browsers on desktop and mobile devices.
- Integration with payment terminals (for in-store pickup or POS systems) may be required.

4.1.3 Software Interfaces

- Integration with user authentication systems for account login, registration, and account management.
- Integration with secure payment gateways to ensure safe online transactions and support various payment methods (credit cards, digital wallets, etc.).
- Integration with third-party services for delivery tracking (if applicable).

4.1.4 Communications Interfaces

- Real-time notifications should be sent to customers regarding order status, shipping, and delivery updates.
- Secure communication between the front-end user interface and the back-end server to ensure data privacy and security.

4.2 Functional Requirements

This section describes specific features of the jewellery shop website. If desired, some requirements may be specified in the use-case format and listed in the Use Cases Section.

4.2.1 Functional Requirement or Feature

4.2.1.1 Product Browsing

- **Introduction**: Users should be able to browse jewellery collections, including rings, necklaces, bracelets, and other categories.
- **Inputs**: User selects a category or searches for a specific item using filters (material, price range, etc.).
- **Processing**: The system retrieves and displays the product listings from the database.
- **Outputs**: User sees product thumbnails, detailed descriptions, and prices.
- **Error Handling**: If the product data fails to load, an error message should be displayed with instructions to refresh the page.

4.2.2 Functional Requirement or Feature

4.2.2 Order Placement

- **Introduction**: Users should be able to select jewellery items, customize them if necessary, and place orders.

- **Inputs**: User selects items, specifies any customizations (e.g., engraving), provides shipping information, and enters payment details.
- **Processing**: The system records the order details, processes the payment, and updates inventory.
- **Outputs**: A confirmation message is displayed with the order summary and estimated delivery time.
- **Error Handling**: If payment fails or an item is unavailable, an appropriate error message should be displayed, and the user should be prompted to choose an alternative payment method or select a different product.

4.2.3 Functional Requirement or Feature
4.2.3 Order Tracking

- **Introduction**: Users should be able to track the status of their order.
- **Inputs**: User logs into their account and selects the "Order History" section to view the status of current and past orders.
- **Processing**: The system retrieves the status of orders from the database and displays relevant details, such as shipment tracking or estimated delivery dates.
- **Outputs**: The user sees the order's current status (e.g., pending, shipped, delivered).
- **Error Handling**: If tracking data is unavailable, a message should indicate that the order status could not be retrieved and suggest the user try again later.

These are just a few functional requirements outlined. The complete list of functional requirements will cover all aspects of the jewellery shop website, including product management, customer interactions, order processes, and error handling mechanisms.

5.1 Non-Functional Requirements

Non-functional requirements are essential to ensure the overall performance, reliability, security, and maintainability of the jewellery shop website. These requirements must be measurable and achievable at a system-wide level.

5.1 Performance

- The system should be able to handle a minimum of 100 concurrent users without significant performance degradation.
- Response times for browsing products, adding items to the cart, and completing transactions should be less than 3 seconds on average.

- The system should be capable of handling peak shopping hours (e.g., holiday sales) without slowdowns or crashes.

5.2 Reliability

- The system should have a mean time between failures (MTBF) of at least 30 days.
- In the event of a failure, the system should be able to recover and resume normal operation within 5 minutes.
- Data integrity should be maintained at all times, with a backup and recovery mechanism in place to prevent data loss.

5.3 Availability

- The system should be available 24/7, with planned maintenance downtime not exceeding 1 hour per week.
- The system should maintain at least 99.9% uptime per month, excluding scheduled maintenance.

5.4 Security

- User authentication and authorization mechanisms should be in place to prevent unauthorized access to sensitive functionalities, such as order history and payment details.
- Payment information should be encrypted during transmission to ensure data security and prevent unauthorized access.
- The system should comply with Payment Card Industry Data Security Standard (PCI DSS) to ensure the protection of payment card information.

5.5 Maintainability

- The codebase should be well-documented, following industry best practices and coding standards.
- Regular code reviews and automated testing should be conducted to ensure code quality and identify potential issues early.
- Updates and patches should be applied promptly to address security vulnerabilities and improve system performance.

5.6 Portability

- The system should be compatible with major web browsers such as Chrome, Firefox, Safari, and Edge.

- The website should be responsive and adaptable to different screen sizes and devices, including desktops, laptops, tablets, and smartphones.
- Compatibility with different operating systems (Windows, macOS, Linux) should be ensured for both users and administrators.

5.7 Design Constraints

Design constraints are factors that limit the options available to the development team when designing the jewellery shop website. These constraints may arise from external standards, company policies, hardware limitations, or other factors.

- **Payment Card Industry Data Security Standard (PCI DSS)**: The system must comply with PCI DSS requirements to ensure the security of payment card information.
- **University Authentication System Integration**: If the website is for a university-based jewellery shop, it may need to integrate with the university's authentication system to allow users to log in using university credentials.
- **Mobile Compatibility**: The system must be compatible with a variety of mobile devices to cater to users who prefer to shop using smartphones or tablets.
- **Hardware Limitations**: The website must account for potential hardware limitations on users' devices, ensuring it operates efficiently even on lower-end smartphones or computers.
- **Data Privacy Regulations**: The system must adhere to data privacy regulations, ensuring that user data (including personal and payment information) is handled and stored securely and in compliance with applicable laws.
- **Third-Party Software Integration**: The system may need to integrate with third-party services for functions such as payment processing, shipping, or inventory management, which may impose design constraints.
- **Scalability**: The system design must be scalable to accommodate future growth in the product catalog, user base, and order volume. The architecture must allow for easy updates and expansions.

5.8 Other Requirements

- **Documentation**: Comprehensive documentation must be provided for system administrators, jewellery shop staff, and end users to ensure smooth deployment, usage, and maintenance of the website.
- **Training**: Training sessions may be required for jewellery shop staff to familiarize them with the system's features, such as inventory management, order processing, and customer service functionalities.
- **Support**: Ongoing technical support and maintenance services may be needed to address any issues or inquiries raised by users or stakeholders post-deployment.

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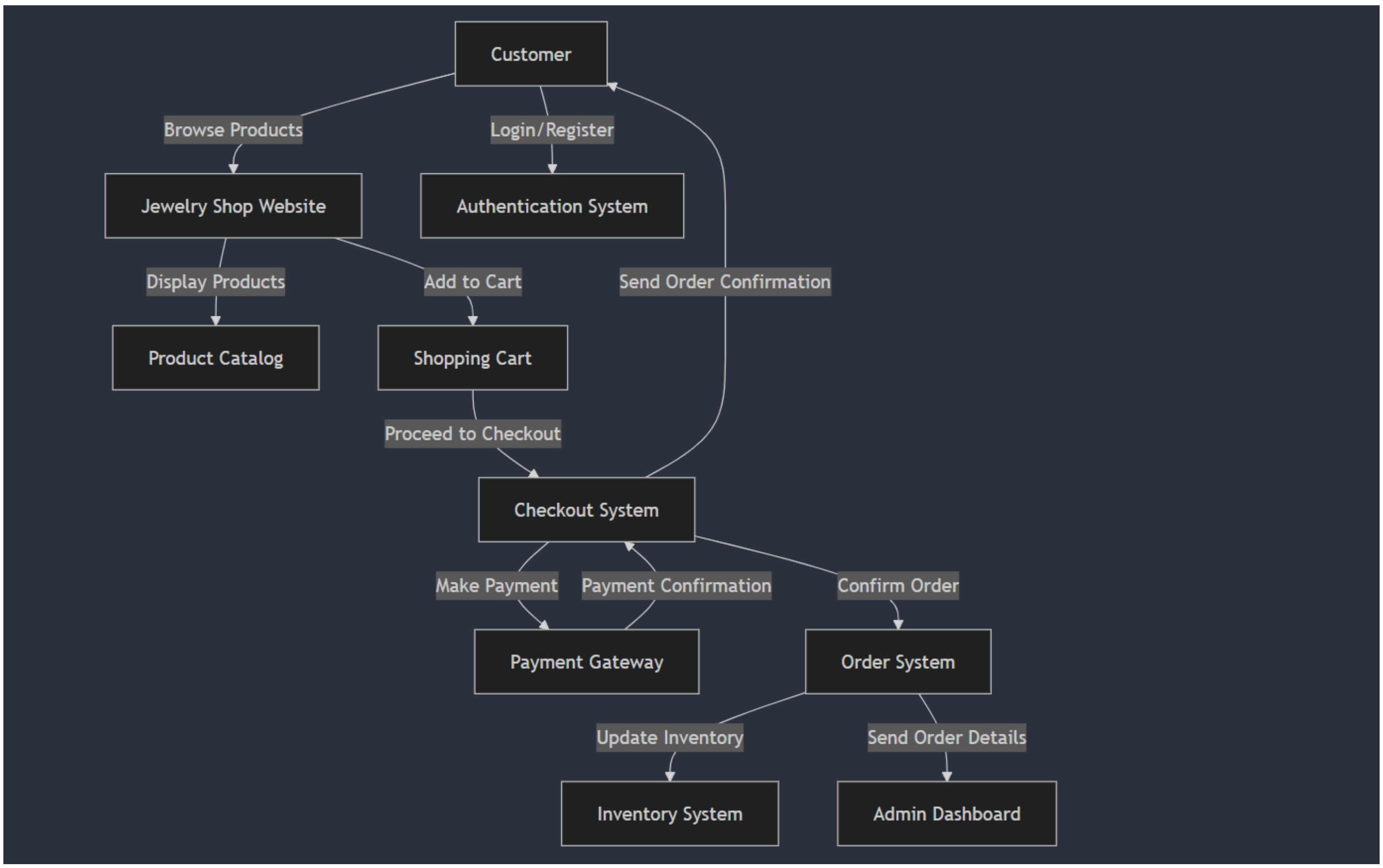
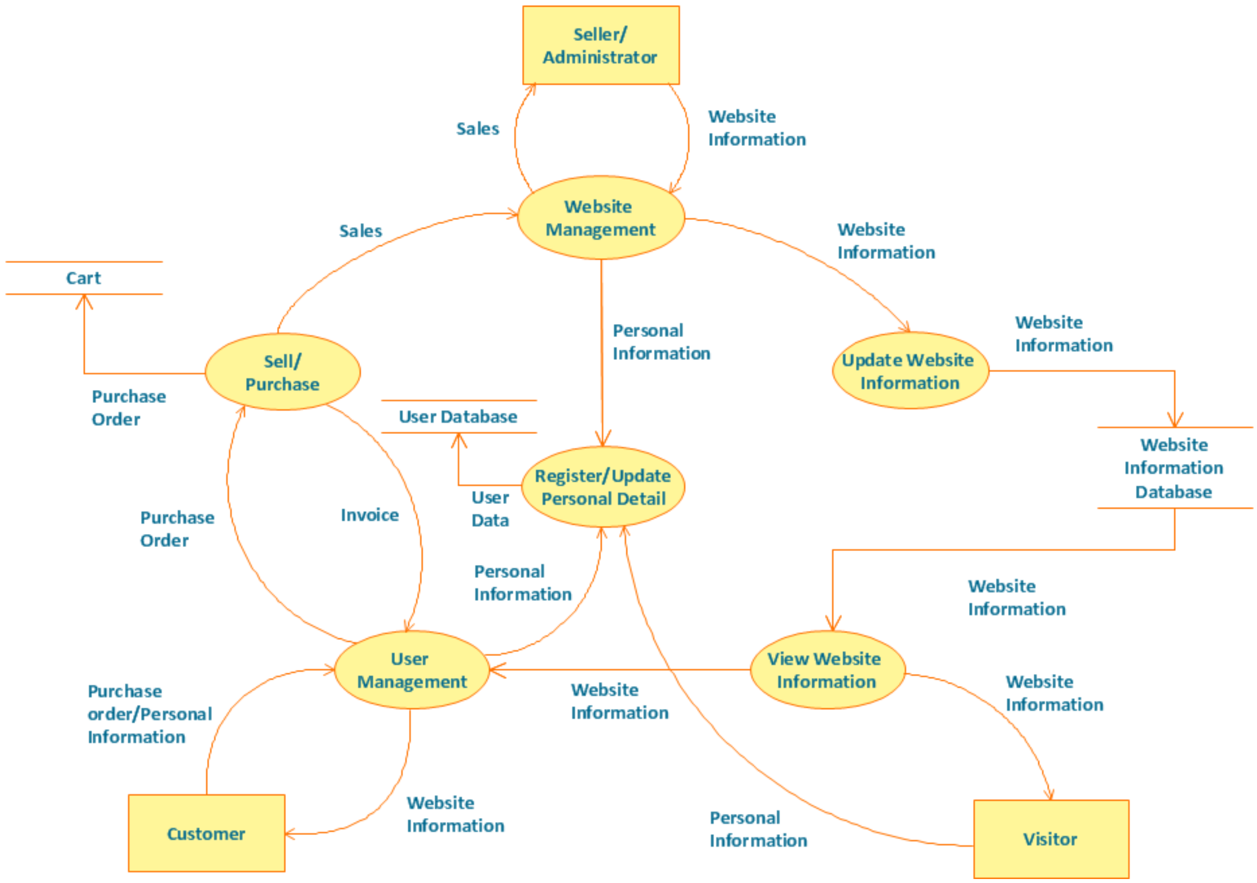
6. Analysis Models

6.1 Data Flow Diagrams (DFD)



6.2 Data Flow Diagram



Video link - https://drive.google.com/drive/folders/1K27oFXwQgu8ZuYlKk69BzwQVUXlvwaTW?usp=drive_link